Machine Learning

08-05-26

ROC- AUC

ROC - Reciever Operating characteristic
AUC - Area under the curve

- ROC- AUC is a performance metric used for binary classifier model

- ROC-AUC is a score betⁿ 0 to 1

ROC AUC score α model performance
higher score = better model

ROC AUC is computed using True positive rate and False positive rate

$$TPR = \frac{TP}{TP+FN}$$ - recall

$$FPR = \frac{FP}{FP+TN}$$

A good classification model should have high TPR and low FPR

y_{act}	1	0	1	0	1
prob pred	0.6	0.2	0.7	0.55	0.3

with threshold ≥ 0.5

y_{pred}	1	0	1	1	0

$acc = \frac{3}{5}$ $\quad prec = \frac{2}{3}$

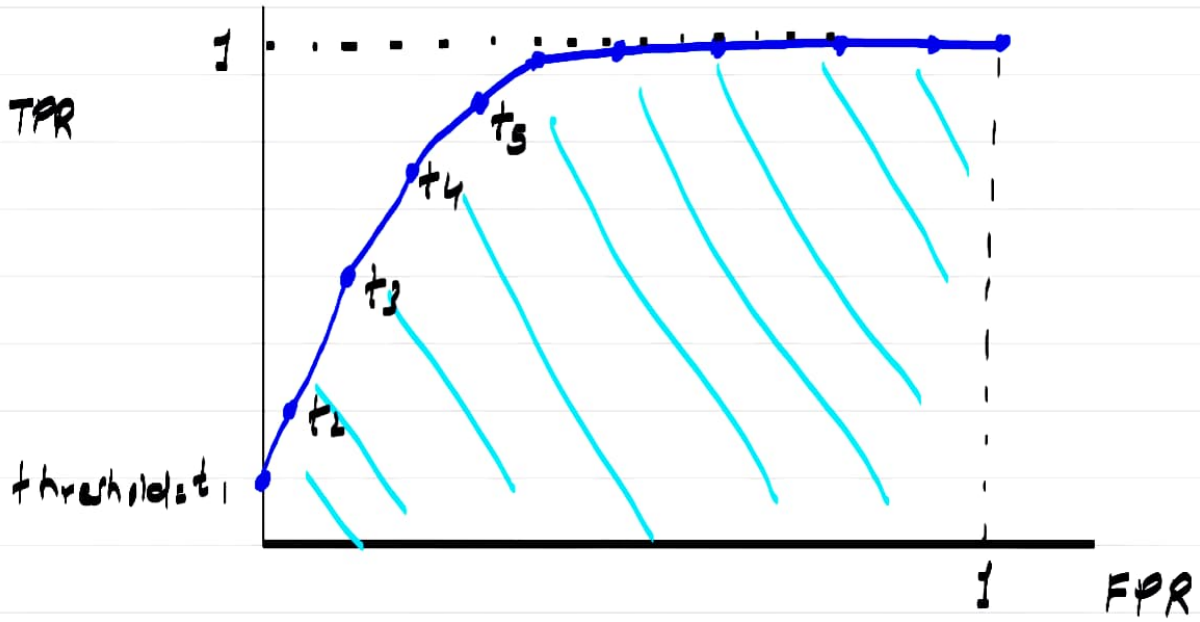
with threshold ≥ 0.6

y_{pred}	1	0	1	0	0

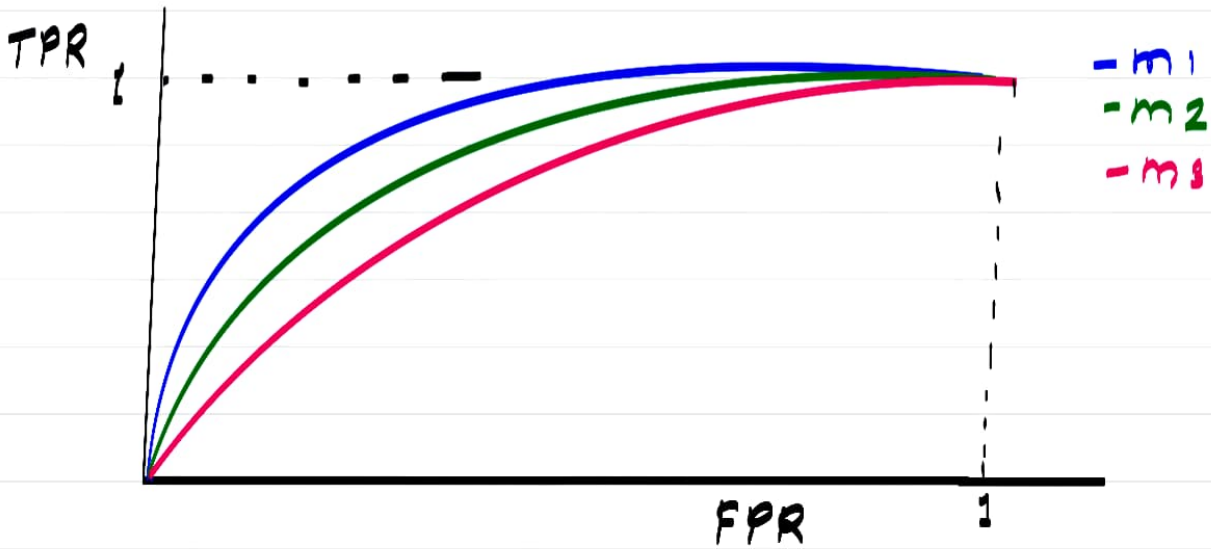
$acc = \frac{4}{5}$ $\quad prec = \frac{2}{2}$

- accuracy, prec, rec, f1 are threshold dependent metrics
- To get overall performance of model over multiple threshold, we use ROC-AUC

- For multiple thresholds we compute TPR and FPR values
- plot them on a graph / curve
- compute area under the curve



ROC-AUC score = area under this curve



area under the curve

$m_1 > m_2 > m_3$

model performance

$m_1 > m_2 > m_3$

ROC-AUC gives overall model performance across multiple thresholds

Ensemble ML models

- In ensemble ML we train multiple models known as weak/base learners and combine them to create a strong learner model

Base/weak learners: multiple small models individually trained

- weak learners learn a small portion of complete task
- these are combined together to get final prediction
- performance of ensemble model is usually better than non-ensemble single models.

Random Forest model >> 1 very big decision tree with 5 DTs

1 linear regr

1 KNN regr

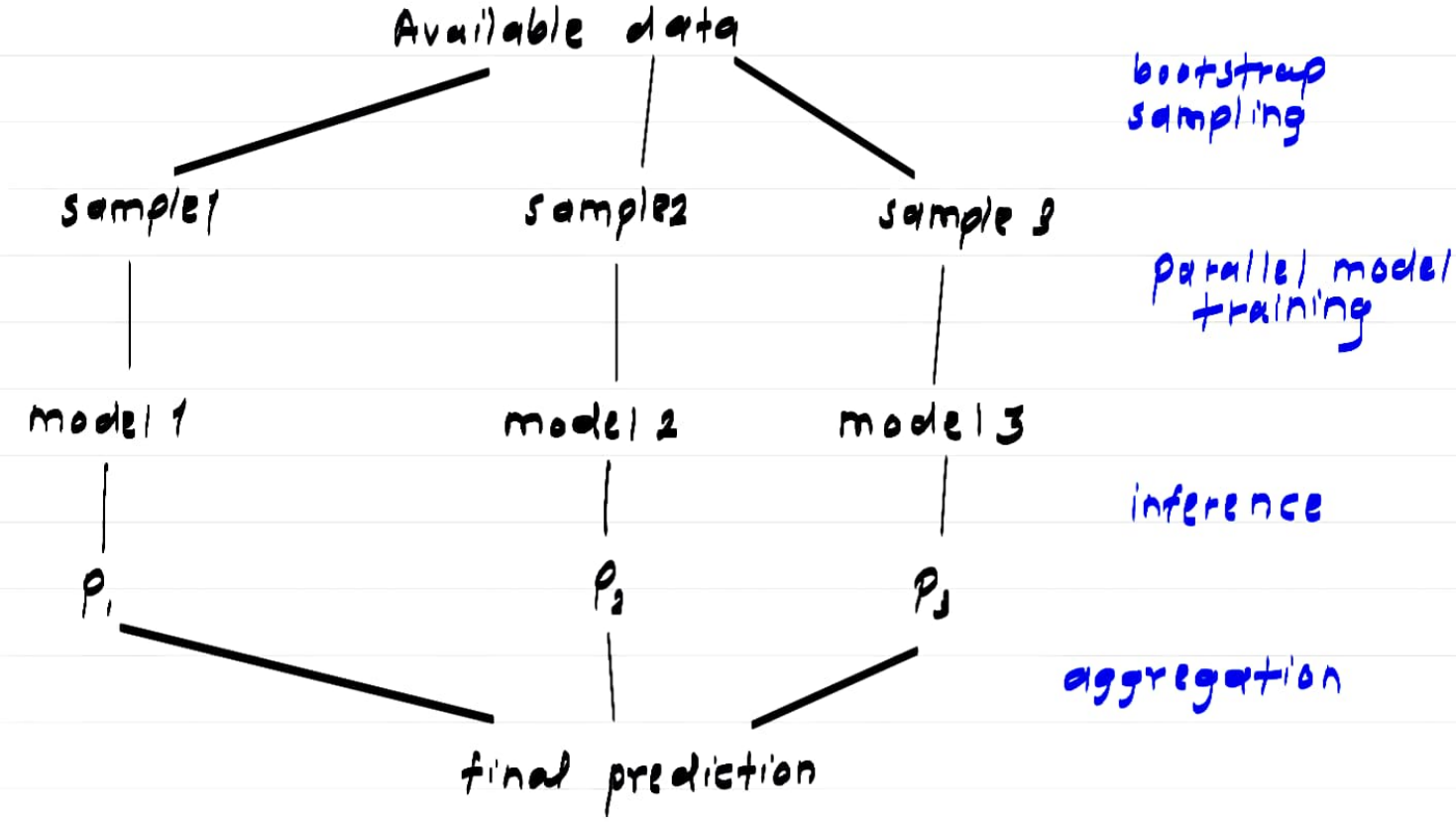
Two major ensemble techniques

1) Bagging - Parallel ensembles

2) Boosting - series ensembles

Parallel Ensembles: Bagging

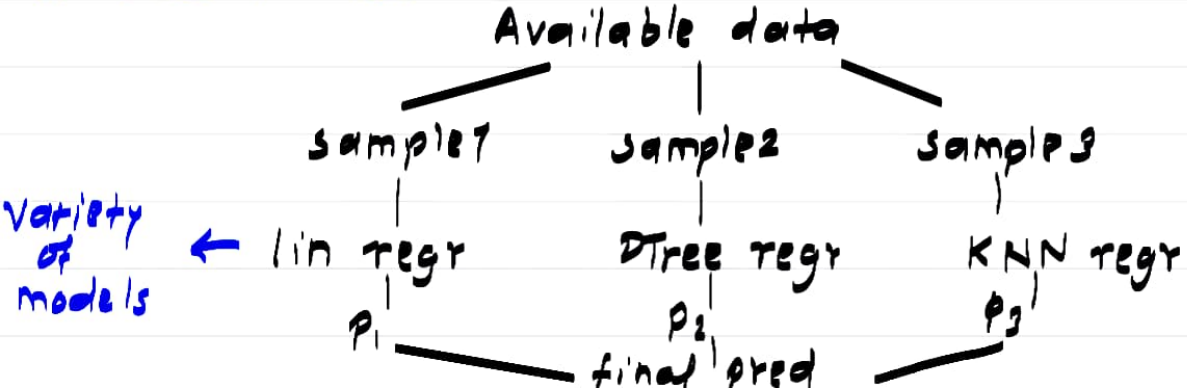
Bootstrap + Aggregation



- In bagging, weak learner models are trained parallely

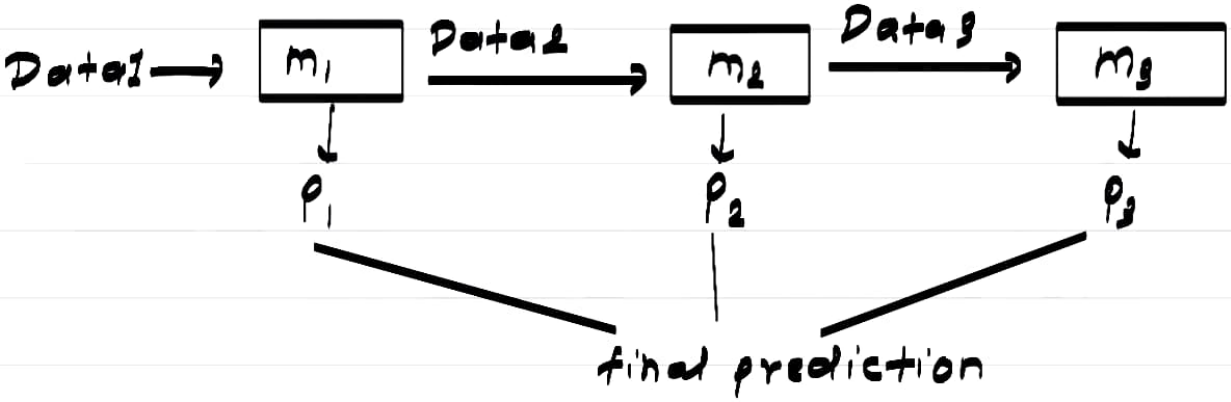
eg. RF - is a bagging ensemble model
- it uses DTrees as base/weak learners

Custom bagging models can be also trained using variety of base learners



Series Ensembles: Boosting

- In boosting, models are trained one after the other in sequence



- every next model in this sequence boosts overall performance of the system by correcting mistakes of the previous model

- Each model in series is a base/weak learner

eg. Adaboost, Gradient boost, XGboost

- Most of the boosting models use decision trees as base learners

Adaboost :

- Adaptive boosting
- used for classification and regression
- trains multiple weak learners sequentially and create a strong model

Adaboost Classifier

	x_1	x_2	x_3	target y
d_1				C_1
d_2				C_1
d_3				C_2
d_4				C_3
d_5				C_2

$X \rightarrow [d_1, d_2, d_3, d_4, d_5]$

1) Decide the no. of base learners to be included in adaboost model

no. of base learners → hyperparameter

- Also decide all the hyperparameters for base learner models
- all base learners are decision trees - all will have same hyperparameters

2) Assign weight to every datapoint

- At the start, we assign equal weight to every datapoint

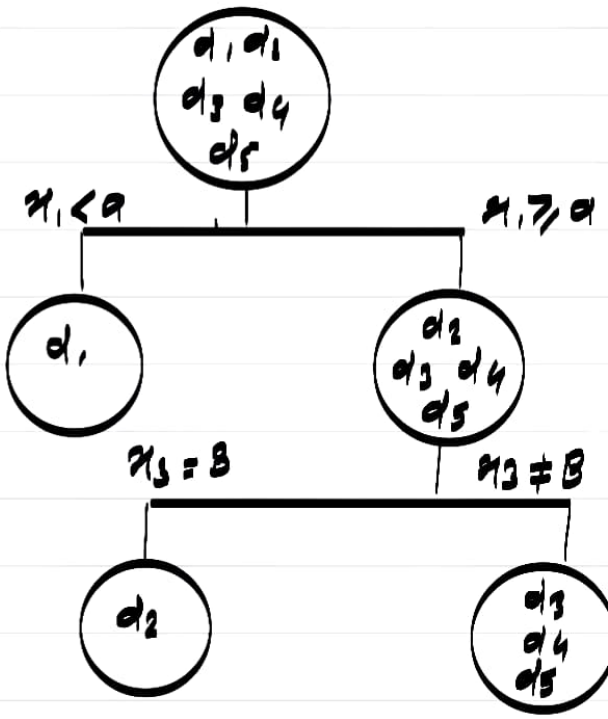
Lets say we decide base learners as DTrees with max depth = 2
min samples split: 2
min samples leaf: 1

3) Train base learner model current data

current data $\longrightarrow X_0$

	x_1	x_2	x_3	target
d_1				c_1
d_2				c_1
d_3				c_2
d_4				c_3
d_5				c_2

model m_1



this is the 1st model in series $\longrightarrow$

	x_1	x_2	x_3	target	$\hat{y}_1$		w_1
d_1				c_1	c_1	✓	0.2
d_2				c_1	c_1	✓	0.2
d_3				c_2	c_2	✓	0.2
d_4				c_3	c_2	✗	0.2
d_5				c_2	c_2	✓	0.2

- model m_1 is making mistake for datapoint d_4

- we want next model m_2 to focus more on d_4 and try to resolve mistake made by m_1

How to make sure that m_2 gives more focus on d_4 ?